

Inferential Statistics: Knowledge Score vs. Employment Status

Independent Two-Sample T-Test

This report examines whether there is a statistically significant difference in clinical knowledge between Employed and Unemployed thalassemia carriers.

Descriptive Statistics

Employment Status	N	Mean (V3 Score)	Std Dev (V3 Score)	Mean (Weighted V3)	Std Dev (Weighted)
Employed	122	10.97	8.53	1.779	4.743
Unemployed	76	11.92	7.50	2.269	4.248

Hypothesis Testing (Welch's T-Test)

We use Welch's T-Test (assuming unequal variances) for robust results.

1. Analysis using the mathematically rigorous Weighted V3 Knowledge Score:

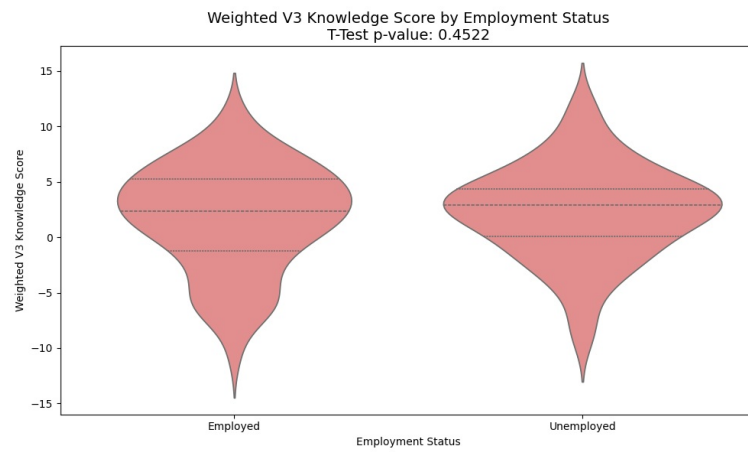
- **T-Statistic:** -0.753
- **P-Value:** 0.4522
- **Conclusion:** Not Statistically Significant ($p > 0.05$). There is no significant difference in knowledge based on employment status.

2. Analysis using the standard V3 Knowledge Score:

- **T-Statistic:** -0.826
- **P-Value:** 0.4102
- **Conclusion:** Not Statistically Significant ($p > 0.05$).

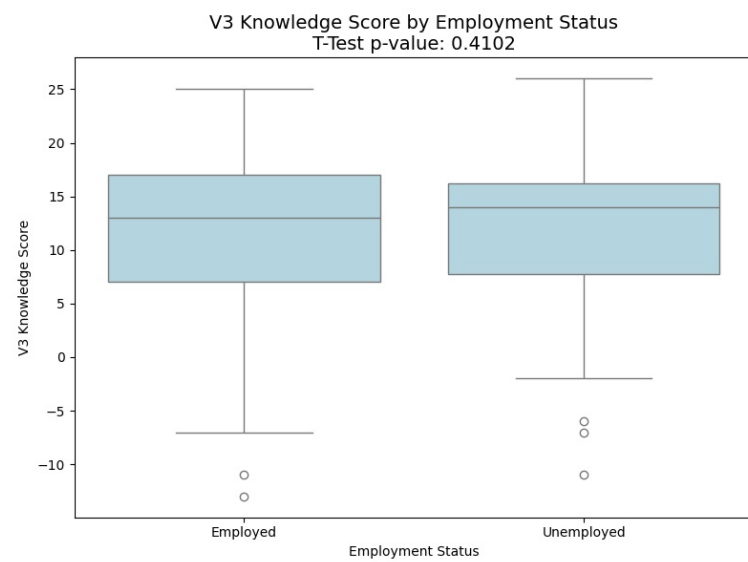
Visualizations

Weighted V3 Score Violin Plot



Violin Plot

Standard V3 Score Box Plot



Box Plot